



PRESIDENCY UNIVERSITY

CSE208: Digital Logic Design Laboratory

Class 03: Half Adder & Half Subtractor

Design, Boolean Expressions, and Truth-Table Verification

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Class 03: Learning Objectives

By the end of today's laboratory, students should be able to:

- 1 Explain the principles of binary addition and subtraction.
- 2 Differentiate between a half adder and a half subtractor.
- 3 Construct the truth tables of both circuits.
- 4 Derive the Boolean expressions for Sum, Carry, Difference, and Borrow.
- 5 Implement a half adder using XOR and AND gates.
- 6 Implement a half subtractor using XOR, NOT, and AND gates.
- 7 Verify both circuits experimentally.

Today's Experiment

Design and implementation of a half adder and a half subtractor.

Class 02: Universal Gates

NAND and NOR $\rightarrow$ Basic Logic Functions

Today's Question

How can we use logic gates to perform **binary arithmetic**?

Binary Inputs



Logic Gates



Arithmetic Outputs

Binary Addition: The Basic Idea

The four possible single-bit addition cases are:

<i>A</i>	<i>B</i>	Result
0	0	0
0	1	1
1	0	1
1	1	10_2

Important Observation

When $1 + 1 = 10_2$, the result contains:

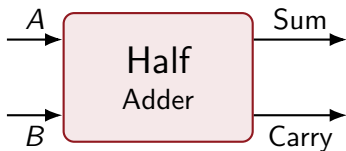
- Sum bit = 0
- Carry bit = 1

A half adder generates these two output bits.

Half Adder: Definition and Operation

Definition

A **half adder** is a combinational logic circuit that adds two single-bit binary inputs and produces a Sum and a Carry.



Inputs

A, B

Outputs

$S = \text{Sum}$

$C = \text{Carry}$

Key Point

The half adder has no carry-in input.

Half Adder: Truth Table

The truth table describes the outputs for every possible combination of inputs.

A	B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Observe

- Sum is HIGH when exactly one input is HIGH.
- Carry is HIGH only when both inputs are HIGH.

Half Adder: Boolean Expressions

1. Sum Output

From the truth table, Sum is 1 for:

$$(A, B) = (0, 1), \quad (1, 0)$$

Therefore,

$$S = \overline{A}B + A\overline{B}$$

This is the XOR operation:

$$S = A \oplus B$$

2. Carry Output

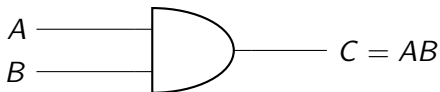
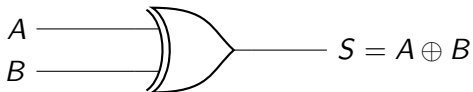
Carry is 1 only when both inputs are 1:

$$C = AB$$

Half Adder: Logic Circuit

The half adder can be implemented using:

- One XOR gate for Sum.
- One AND gate for Carry.



Implementation

Both gates receive the same inputs, A and B . Their outputs are Sum and Carry, respectively.

Binary Subtraction: The Basic Idea

A half subtractor performs:

$$A - B$$

The four possible cases are:

A	B	Result
0	0	0
0	1	-1
1	0	1
1	1	0

Important Observation

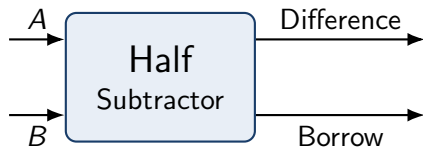
When $A = 0$ and $B = 1$, a borrow is required from the next higher bit.

The circuit generates a Difference bit and a Borrow bit.

Half Subtractor: Definition and Operation

Definition

A **half subtractor** is a combinational logic circuit that subtracts one single-bit binary input from another and produces a Difference and a Borrow.



Inputs

A, B

Outputs

$D = \text{Difference}$

$B_{\text{out}} = \text{Borrow}$

Operation

Subtract input B from input A .

Half Subtractor: Truth Table

A	B	Difference (D)	Borrow (B_{out})
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Observe

- Difference is 1 when the inputs are different.
- Borrow is 1 only when $A = 0$ and $B = 1$.

Half Subtractor: Boolean Expressions

1. Difference Output

Difference is 1 when the inputs are different.

$$D = \overline{A}B + A\overline{B}$$

Therefore,

$$D = A \oplus B$$

2. Borrow Output

Borrow occurs only when $A = 0$ and $B = 1$.

$$B_{\text{out}} = \overline{A}B$$

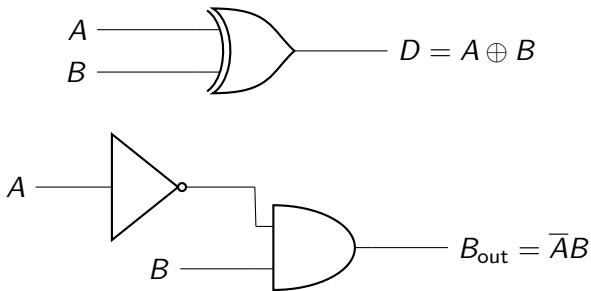
Final Half Subtractor Equations

$$D = A \oplus B, \quad B_{\text{out}} = \overline{A}B$$

Half Subtractor: Logic Circuit

The half subtractor can be implemented using:

- One XOR gate for Difference.
- One NOT gate to obtain $\bar{A}$.
- One AND gate for Borrow.



Half Adder vs. Half Subtractor

Feature	Half Adder	Half Subtractor
Operation	$A + B$	$A - B$
Inputs	A, B	A, B
First output	Sum: $A \oplus B$	Difference: $A \oplus B$
Second output	Carry: AB	Borrow: $\overline{A}B$
Special case	Carry when $A = B = 1$	Borrow when $A = 0, B = 1$

Key Observation

Both circuits use XOR for their first output. Their second outputs distinguish addition from subtraction.

Laboratory Procedure

Part A: Half Adder

- 1 Identify the XOR and AND gate ICs.
- 2 Connect the IC power supply correctly.
- 3 Construct the circuit using the circuit diagram.
- 4 Apply all four input combinations.
- 5 Record Sum and Carry.

Part B: Half Subtractor

- 1 Identify the XOR, NOT, and AND gate ICs.
- 2 Construct the circuit.
- 3 Apply all four input combinations.
- 4 Record Difference and Borrow.

Safety and Troubleshooting

Check VCC, GND, IC orientation, and input connections before applying power. Do not leave logic inputs floating.

Experimental Observation

Students should complete the following tables during the laboratory experiment.

Half Adder

<i>A</i>	<i>B</i>	<i>S</i>	<i>C</i>
0	0		
0	1		
1	0		
1	1		

Half Subtractor

<i>A</i>	<i>B</i>	<i>D</i>	<i>B</i> _{out}
0	0		
0	1		
1	0		
1	1		

Verification

Compare the measured outputs with the theoretical truth tables and identify any discrepancies.

Viva Questions

- ① What is a half adder?
- ② Why is it called a half adder?
- ③ What are the outputs of a half adder?
- ④ Why is XOR used to generate the Sum?
- ⑤ Under what condition does a half adder generate Carry?
- ⑥ What is a half subtractor?
- ⑦ When does a half subtractor generate Borrow?
- ⑧ Why is an inverter required in the Borrow circuit?
- ⑨ What is the difference between Carry and Borrow?
- ⑩ Can a half adder add three input bits?

Class Summary

Half Adder

$$S = A \oplus B$$

$$C = AB$$

- Performs binary addition.
- Produces Sum and Carry.
- Uses XOR and AND gates.

Half Subtractor

$$D = A \oplus B$$

$$B_{\text{out}} = \bar{A}B$$

- Performs binary subtraction.
- Produces Difference and Borrow.
- Uses XOR, NOT, and AND gates.

Next Class: Full Adder

Adding three input bits using Carry-in.